Adaptability — Implementation

What Is Built, What Has Run, and the One Place They Do Not Meet

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DRAFT — draft-2026-08-30.4 r4 (1 September 2026). Circulated for comment, not as a finished result. The companion states the theory; this paper reports the implementation and its principal gap. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

The operation the companion describes is built, has run at scale, and has been measured, including a large correction against itself. It is also, in one specific and consequential way, in the wrong place. *The algebra is native; the faculty that uses it is not.* Every count below was obtained by running a command against the tree.

1 What is built

The transform algebra is native to the engine. Anti-unification retaining its witness, substitution, unification, structural similarity and a coverage predicate are all term-level operations, evaluable inside the rule language rather than beside it. The shape ladder is built on them: decoding against a held codebook, transformation within the held basis, inference of a dimension, and minting.

The ladder runs as rules. It is a rule fixpoint rather than a loop in the host language — a residual mints a quantum, a codebook epoch re-arms decoding, and the re-decode either recognises or does not. That much is graph-resident and gated.

2 What has run

Remark 2.1 (Transfer by parallelogram completion). The strongest demonstration takes four of the agent's own rules arranged as a 2×2 over two independent dimensions. *Parallelogram* is borrowed from the embedding-arithmetic literature and the companion sets out the lineage, along with the reason that literature's known critique does not carry over to an exact symbolic operation. Anti-unification over two corners derives the transform; substitution applies it to the third; the result is a rule *absent from all* 2,851 *distinct recognisers in the corpus* which, installed, fires correctly on the missing cell.

318 such families exist in the corpus, and 200 of 200 sampled transfers are algebraically lossless, well-defined and novel. *No new term variants, no graph mirror and no engine change were required to demonstrate this* — the capability was reachable with what already existed.

Remark 2.2 (The concept-grain faculty, and its own correction). At the concept grain the faculty has run against the agent's largest real bridge, 142 concepts against 137, and reported 8,575 analogies.

Of those, 8,253 *were an artefact*: a keyword matching *inside* a longer word, which accounted for 39.6% of every vocabulary hit in the booted graph. After token-anchoring and a requirement of structural support the count fell to 321, of which 78 are corroborated by a witness shared across pairs.

That is reported here as the faculty's principal measured result, not as an embarrassment adjacent to one. A ninety-six per cent retraction is what the correction was worth, and the corrected number is the one this paper stands behind.

3 The gap

The two halves above do not meet, and the seam is exactly locatable.

Anti-unification with witness retained appears **seven times** in the faculty's host-language module and **zero times** across the 3,843 rules carried by the 805 seed manifests. The faculty is 1,241 lines of host code, and it *does* call the native algebra — so this is not a capability that was built and left unused. It has run, it has been measured, and its output survives correction.

What it is not is graph-resident. The transform algebra is reachable from the host language and not from a rule.

Remark 3.1 (Why that matters here more than anywhere else)*.* This programme's boundary on self-repair is sharp and is stated elsewhere in it: *the agent can repair and extend its cognition exactly where that cognition is graph-native*, because what it can vary, inspect and mint against is graph data. A faculty in the host language can be used and cannot be improved by the thing using it.

So of all the faculties that could have been left outside, *this is the worst one* — because it is the adaptation mechanism. The system's ability to acquire competences it was not given is itself a competence it cannot acquire. That is not a contradiction, but it is a ceiling, and it is the ceiling the companion's ladder was built to remove.

The measured retraction above sharpens it. A ninety-six per cent artefact rate was found by an audit performed *on* the faculty from outside. A graph-resident faculty is the kind of thing this substrate has repeatedly been made to notice about itself — it files gaps, retires its own duplicates, and surfaces what it cannot answer. *The faculty most in need of that treatment is the one that cannot receive it.*

4 What the gap is not

It is not that the algebra is unreachable in principle. Neighbouring operations from the same family are already used in rules — coverage in twelve seed files, unification in eleven, substitution in three. The vocabulary is live; the one operation that carries the transform is the one no rule invokes.

It is not that the demonstration is unsound. The parallelogram result stands on its own terms, and its numbers are reported above with the caveat they deserve: it was demonstrated in analysis scripts rather than as agent behaviour, so it establishes that the corpus *contains* 318 derivable rules, not that the agent derived them.

And it is not a large amount of work in prospect. The absence of new term variants in the demonstration is the relevant measurement: what would be required is a rule surface over an algebra that already evaluates, which is the same move this programme has made before and reported.

5 The contrast the record supplies

One further measurement belongs here, because it is uncomfortable in the right direction.

Program synthesis in this substrate is *graph-native* in the way analogy is not: its enumerator is a term, and the decisions — propose, evaluate, accept — are seed rules. And its clearest result about held content is a null. Running the loop with the acquired content and running it stripped produced 54 of 257 in both arms, *on the identical set of tasks*, with the difference measured at zero repeatedly.

So the record contains both of these at once. *The faculty that is graph-native produced a clean null about the value of what it held; the faculty that is not produced the strongest positive result in the corpus, in analysis scripts.* Neither observation rescues the other, and the pair is reported together because taking either alone would misrepresent the record — the first is not a reason to doubt the second, and the second is not a reason to discount the first.

The reading that survives both is narrow, and the companion states its general form: capability has plural sources selected by regime, so which faculty happens to be graph-resident says nothing about which one carries a given domain. *Being graph-resident is what makes a faculty improvable by the system carrying it, and it is not what makes a faculty work.* The gap in Section 3 is a ceiling on self-repair, and this section is the reminder that removing it would not by itself buy a capability.

6 Falsifiability commitments

1. [DETAIL] **The transform algebra is absent from the rule layer.** Fails on the exhibition of any seed rule invoking witness-retaining anti-unification; the count is zero across 3,843 rules at the stated revision and is a command, not a judgement.

2. [SHAPE] **Self-repair is bounded by graph-nativeness.** Fails if the agent can be shown to improve a host-language faculty by its own operation rather than by an author's — which would remove the ceiling this paper reports and much of its point.

3. [DETAIL] **The** 318 **families are derivable and novel.** Fails if a sampled transfer is found to be lossy, ill-defined, or to reproduce a recogniser already in the corpus.

4. [SHAPE] **The parallelogram result is about the corpus, not the agent.** This paper commits to the negative: no claim is made that the agent derived those rules, and a reader who extracts one has found an error rather than a result.

5. [DETAIL] **The corrected analogy count is the defensible one.** Fails if token-anchoring and the structural-support requirement can be shown to discard genuine correspondences rather than artefacts, which would make 321 an over-correction rather than a correction.